Time and Tide Revision

Q1 Define the following,

ellipse _____

tide _____

penumbra _____

umbra _____

neap tide _____

Q2 Which statements are true?

a) The moon is luminous. _____

b) The Sun in luminous. _____

c) The Sun is a star. _____

d) The shadow of the Earth on the moon is called a solar eclipse. _____

e) The moon is a satellite. _____

f) The Earth rotates around the sun in a clockwise direction. _____

g) The sun appears to move across the sky from east to west. _____

Q3 How long does it take for:

a) the earth to rotate on its axis? _____

b) the earth to revolve around the sun? _____

c) the moon to rotate on it axis? _____

d) the moon to revolve once around the earth and line up with the same star again? _____

e) the moon to revolve once around the earth and line up again with the sun and the earth? _____

Q4 Answer the following in your pad.

a) Why do day and night occur? Explain your answer and include a **labelled** diagram.

b) What causes high and low tides?

c) What conditions cause spring tides and neap tides? Explain your answer and draw an **annotated** diagram.

Q5 In your pad:

Explain the differences between a lunar and solar eclipse. Consider their differences in: positions of Sun, Moon and Earth. Include a labelled diagram for both eclipses.

Q6 Moon Phases

Draw a series of eight pictures in the correct sequence showing the phases of the moon in the **southern** hemisphere.

Q7 Graphing

The following data shows the tide height for high tide at different phases of the moon during the month of May.

Moon Phase	New Moon	Waxing Crescent	First Quarter	Waxing Gibbous	Full Moon	Waning Gibbous	Last Quarter	Waning Crescent
Date	3 May	8 May	10 May	13 May	17 May	22 May	24 May	27 May
Height	2.7	2.4	2.5	2.7	3.2	2.6	2.4	2.3

Use this data to draw a line graph. Make sure you include all elements of a good graph – a title, labelled x and y axes and an appropriate scale. Join the points with a smooth curved line.

